

ROGELIO RUZCKO TOBIAS

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EDUCATION

PhD in Artificial Intelligence

2023~

Ulsan National Institute of Science and Technology (UNIST)

Ulsan, South Korea

- Global Korea Scholarship

MS in Electronics and Communications Engineering

September 2021

De La Salle University

Manila, Philippines

- CGPA: **4.00 / 4.00**
- Research Award

BS in Electronics Engineering

September 2017

Asia Pacific College

Makati, Philippines

- SM Foundation Scholarship

Korean Language and Culture Studies

August 2023

Sunchon National University

Suncheon, South Korea

- Global Korea Scholarship
- TOPIK II Level 3 (한국어능력시험 3급)

Special Science Curriculum Secondary Education

March 2012

Manila Science High School

Manila, Philippines

RESEARCH AND ACADEMIC EXPERIENCE

Researcher, Climate Intelligence Lab

2026 – Present

DI Lab, Inc.

South Korea

- Develop AI models and software for satellite-based climate and disaster monitoring, AI-Radar Map enhancement, and precipitation forecasting
- Fuse satellite imagery, radar, ground observations, numerical weather prediction data, terrain, and environmental data to support climate-risk decisions in agriculture and water management

Researcher, Intelligent Remote Sensing and

2022 – 2026

Geospatial Information Science Laboratory

Ulsan National Institute of Science and Technology (UNIST)

Ulsan, South Korea

- Developed AI methods for satellite imagery, remote sensing, and geospatial analysis, including change detection, soil-moisture retrieval, disaster assessment, and precipitation nowcasting
- Contributed to a Korean government-funded project supporting AI algorithm development for the GEO-KOMPSAT-5 (GK-5) satellite
- Won the 2024 Grand Prize in Algorithm Development at the Satellite Information Application Contest organized by the Korea AeroSpace Administration (KASA)

Assistant Professor, College of Engineering

2021 – 2022

National University

Manila, Philippines

- Published 3 multidisciplinary research papers with the Office of the Vice President for Research and Development focusing on the applications of Artificial Intelligence and Machine Learning
- Coordinated strategic actions to enhance the research index of National University through faculty research collaborations
- Member, National University Research Council

MS Student Researcher, Intelligent Systems Laboratory

2019 – 2021

De La Salle University

Manila, Philippines

- Published 4 journal articles and 15 research papers by collaborating with the DLSU Intelligent Systems Laboratory focused on precision agriculture, business intelligence, and biomedical engineering

Lecturer, School of Engineering

2018 – 2020

Asia Pacific College

Makati, Philippines

- Published 6 multidisciplinary research papers focusing on image processing, neural networks, machine learning, and app development
- Subjects handled: Computer Programming, Engineering Mathematics, Engineering Management, Statistics, and Control Systems
- Trained and mentored students for national competitions, won 6 awards from 2018 to 2019
- Best Project, Design Thinking for the Classroom

ADMINISTRATIVE EXPERIENCE

Multi-Campus Coordinator, Quality Management Office

2021 – 2022

National University

Manila, Philippines

- Co-chair for the Quacquarelli Symonds (QS) Stars Rating committee
- Developed and implemented strategic initiatives that successfully enhanced institutional standards and practices, leading to the achievement of **3 QS Stars rating**
- Created strategic plans and actions to establish National University in local and international rankings
- Maintained the quality standards, services, and policies across campuses, ensuring compliance with the government and accrediting institutions (CHED, ISO, PACUCOA, PTC-ACBET, & PICAB)
- Top 1, Outstanding Performance Awardee

INDUSTRY EXPERIENCE

Consultant, Data Science and Analytics

2022

Qadworks Technical Solutions, Inc.

USA

- Advanced analytics and insights generation: transformed complex data sets into actionable insights, utilizing advanced analytics techniques and tools

Consultant, Technology Solutions

2021

Talkpush, Ltd.

Hong Kong

- Collaborated with the Conversation Design and Engineering team on enhancing chatbots and mobile apps, and developed web applications, translating UI/UX designs into functional code

Developer Intern, Robotic Process Automation

2017

Accenture, Inc.

Philippines

- Developed web-based applications to automate the collection, transformation, and delivery of a company's operating data
- Best Automation Project, Automation University

Intern, IT Shared Services

2016

SM Investments Corporation

Philippines

- Maintained IT infrastructure, managed help desk issues, and monitored the network of the SM group of companies

PROFESSIONAL LICENCES

Electronics Engineer

2018 – Present

Professional License Number: 0069694

Electronics Technician

2018 – Present

Professional License Number: 0016384

TECHNICAL SKILLS

AI Research Areas: Deep Learning, Computer Vision, Remote Sensing, AI Agents, and Language Models

Programming Languages: Python, SQL, R, C++, MATLAB

ML/DL Frameworks: PyTorch, TensorFlow, scikit-learn

Remote Sensing and Geospatial: Change Detection, Geo-LLM, Forecasting, and Super-Resolution

Data Analysis and Visualization: pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI

Research and Development Tools: Git, GitHub, Jupyter, Google Colab, Visual Studio Code, $\LaTeX$

Cloud Platforms: Microsoft Azure, Google Cloud Platform

AFFILIATIONS

PIKO (*Pinoy Iskolars sa Korea*) · Member

NRCP (*National Research Council of the Philippines*) · Associate Member

IEEE (*Institute of Electrical and Electronics Engineers*) · Member

IECEP (*Institute of Electronics Engineers of the Philippines*) · Member

IRCA (*International Register of Certificated Auditors*) · Lead Auditor - ISO 9001

CQI (*Chartered Quality Institute*) · Lead Auditor - ISO 9001

PACUCOA (*Philippine Association of Colleges and Universities Commission on Accreditation*) · Accreditor

Office of the Vice President for Research and Development · National University

- Springer WorldS4 2022 · London, United Kingdom
- TENCON 2021 · Auckland, New Zealand
- IALP 2021 · Singapore, Singapore

Office of the Vice Chancellor for Research and Innovation · De La Salle University

- LifeTech 2020 · Kyoto, Japan

Research Conference Grant · Asia Pacific College

- R10 HTC 2020 · Kuching, Malaysia
- TENCON 2020 · Osaka, Japan
- iC3I 2019 · Singapore, Singapore
- HNICEM 2019 · Ilocos Norte, Philippines

PUBLICATIONS

Research Profiles

-  [Google Scholar](#)
-  [ResearchGate](#)
-  [ORCID](#) · 0000-0003-3353-6074

Journal Publications

- Lee, J., Park, H., Im, J., Koyama, C., **Tobias, R. R.**, & Tadono, T. (2026). ESCAPE: An ensemble-based self-calibrated autoencoder with physics-informed estimation of high-resolution soil moisture and surface roughness from ALOS-2/PALSAR-2 polarimetric observations. *International Journal of Applied Earth Observation and Geoinformation*, 147, 105206.
- **Tobias, R. R.**, Bae, S., Cho, H., & Im, J. (2024). SAMBA: Synthetic data-augmented Mamba-based change detection algorithm using KOMPSAT-3A imagery. *Korean Journal of Remote Sensing*, 40(6-3), 1505–1521.
- **Tobias, R. R.**, Minglana, J., Hernandez, D. K., Mital, M. E., Roxas, R. E. (2023). Artificial Intelligence Applications in Quality Management Systems of Philippine Higher Education Institutions. Springer WorldS4.
- Lauguico, S. C., Concepcion, R. S., Alejandrino, J. D., **Tobias, R. R.**, Macasaet, D. D., and Dadios, E. P. (2020). A Comparative Analysis of Machine Learning Algorithms Modeled from Machine Vision-Based Lettuce Growth Stage Classification in Smart Aquaponics. *International Journal of Environmental Science and Development*.
- Lauguico, S. C., Concepcion, R. S., Alejandrino, J. D., **Tobias, R. R.**, and Dadios, E. P., “Lettuce life stage classification from texture attributes using machine learning estimators and feature selection processes,” *Int. J. Adv. Intell. Informatics*, vol. 6, no. 2, p. 173, 2020.
- Alejandrino, J. D., Concepcion, R. S., Lauguico, S. C., **Tobias, R. R.**, Venancio, L., Macasaet, D., Bandala, A. A., and Dadios, E. P. (2020). A Machine Learning Approach of Lattice Infill Pattern for Increasing Material Efficiency in Additive Manufacturing Processes. *International Journal of Mechanical Engineering and Robotics Research*, 1253–1263. <https://doi.org/10.18178/ijmerr.9.9.1253-1263>
- Concepcion, R. S., Alejandrino, J. D., Lauguico, S. C., **Tobias, R. R.**, Sybingco, E., Dadios, E. P., and Bandala, A. A. (2020), “Lettuce growth stage identification based on phytomorphological variations using coupled color superpixels and multifold watershed transformation,” *Int. J. Adv. Intell. Informatics*, vol. 6, no. 3, p. 261, 2020.

Conference Publications

- **Tobias, R. R.**, Cho, H., & Im, J. (2025, August). Building Damage Assessment Captioning: A Vision-Language Dataset Leveraging State Space Models. In *IGARSS 2025 IEEE International Geoscience and Remote Sensing Symposium* (pp. 1445–1449). IEEE.
- **Tobias, R. R.** et al., “Faster R-CNN Model with Momentum Optimizer for RBC and WBC Variants Classification,” LifeTech 2020 - 2020 IEEE 2nd Global Conference on Life Sciences and Technologies, Mar. 2020, pp. 235–239.
- **Tobias, R. R.** et al., “CNN-based Deep Learning Model for Chest X-ray Health Classification Using TensorFlow,” 2020 RIVF International Conference on Computing and Communication Technologies, pp. 1–6, 2020.
- Mital, M. E., **Tobias, R. R.**, et al., “Transfer Learning Approach for the Classification of Conidial Fungi (Genus *Aspergillus*) Thru Pre-trained Deep Learning Models,” 2020 IEEE Reg. 10 Conf., pp. 1–6, 2020.
- Guillermo, M., **Tobias, R. R.** et al., “Detection and Classification of Public Security Threats in the Philippines Using Neural Networks,” LifeTech 2020 - 2020 IEEE 2nd Global Conference on Life Sciences and Technologies, Mar. 2020, pp. 320–324.
- Concepcion, R. S., Lauguico, S. C., **Tobias, R. R.**, Dadios, E. P., Bandala, A. A., and Sybingco, E. (2020). “Estimation of Photosynthetic Growth Signature at the Canopy Scale Using New Genetic Algorithm-Modified Visible Band Triangular Greenness Index.” 2020 International Conference on Advanced Robotics and Intelligent Systems (ARIS), Taipei, Taiwan, pp. 1–6.

2025 KAGIS Fall Conference and IAGi International Symposium

- Hybrid Deterministic-Diffusion Framework for Probabilistic Precipitation Nowcasting
- Jeju Island, South Korea, November 2025

IGARSS 2025 · 2025 IEEE International Geoscience and Remote Sensing Symposium

- Building Damage Assessment Captioning: A Vision-Language Dataset Leveraging State Space Models
- Brisbane, Australia, August 2025

KMS 2025 · 2025 Korean Meteorological Society Spring Conference

- HYDRO: Hybrid Deterministic-Residual Diffusion Framework for Precipitation Nowcasting
- Seoul, South Korea, May 2025

IALP 2021 · 2021 International Conference on Asian Language Processing

- Scientific Landscape of Publications in Natural Language Processing in the ASEAN Region on COVID-19: A Bibliometric Approach
- Singapore, Singapore, December 11 to 13, 2021

TENCON 2021 · The 2021 IEEE Region 10 Conference

- Science Mapping of Social Media Analytics in Health Through Artificial Intelligence
- Auckland, New Zealand, December 07 to 10, 2021

TENCON 2020 · The 2020 IEEE Region 10 Conference

- Hybrid Tree-Fuzzy Logic for Aquaponic Lettuce Growth Stage Classification Based on Canopy Texture Descriptors
- Osaka, Japan, November 16 to 19, 2020

LifeTech 2020 · The 2020 IEEE 2nd Global Conference on Life Sciences and Technologies

- Android Application for Chest X-ray Health Classification from a CNN Deep Learning TensorFlow Model
- Faster R-CNN Model with Momentum Optimizer for RBC and WBC Variants Classification
- Mielparque Kyoto, Kyoto, Japan, March 10 to 12, 2020

RIVF 2020 · 2020 RIVF International Conference on Computing and Communication Technologies

- CNN-based Deep Learning Model for Chest X-ray Health Classification Using TensorFlow
- RMIT University, Ho Chi Minh City, Vietnam, April 6 to 7, 2020

HNICEM 2020 · Humanoid, Nanotechnology, Information Technology, Communication, Environment, and Management

- Design and Construction of a Solar Energy Module for Optimizing Solar Energy Efficiency
- Philippines, December 28 to 29, 2020

iC3I 2019 · The Fourth International Conference on Contemporary Computing and informatics

- Throat Detection and Health Classification Using Neural Network
- Amity Global Institute, Singapore, December 12 to 14, 2019

HNICEM 2019 · Humanoid, Nanotechnology, Information Technology, Communication, Environment, and Management

- Classification of Landcover from Combined LiDAR and Orthophotos Using Support Vector Machine
- Fort Ilocandia Resort Hotel, Laoag City, Ilocos Norte, Philippines, November 28 to December 1, 2019

PROFESSIONAL AND ACADEMIC ENGAGEMENTS

Scholarly Reviewer · Selected

- International Journal of Imaging Systems and Technology (IMA)
- The IEEE International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM 2020 – 2021)
- The 2021 IEEE Region 10 Conference (TENCON 2021)
- The 2021 IEEE/SICE International Symposium on System Integration (SII 2021)
- The 2021 International Conference on Robotics and Automation (ICRA 2021)

Resource Speaker

- Intelligent Green Box (IGB) Capacity Training · Eastern Visayas State University, Tacloban City, March 03, 2021

Technical Committee Member

- The IEEE International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM 2019 – 2021)

Adviser

- Student Organization Association of Regents (APC Student Council) · Asia Pacific College, 2019 - 2020

Mentor/Coach

- 3rd runner-up, Synergy: Electrical and Electronics Engineering Summit 2019 · UP Diliman, Quezon City, Philippines
- 2nd Runner-up, Emerson Student Innovation 2019 · Innovation Center, Miriam College, Quezon City, Philippines
- 1st Runner-up, 15th Sweep Innovation and Excellence Awards · Makati, Philippines
- Top 10 Finalist, Accenture Program the Future 2019 · Bonifacio Global City, Taguig, Philippines

AWARDS AND RECOGNITIONS

Gilas Award · Embassy of the Republic of the Philippines to the Republic of Korea, South Korea, 2023, 2025, and 2026
Outstanding Paper Award · The Korean Association of Geographic Information Studies and the International Association of Geo-informatics, South Korea, 2025
Grand Prize - Algorithm Development · Korea Aerospace Administration (KASA), South Korea, 2024
Research Award · DLSU Office of the Vice Chancellor for Research and Innovation, 2021
Outstanding Performance Award · National University, 2021
Best Design Thinking Project · Asia Pacific College, 2019
Best Automation Project · Accenture, Inc. Automation University, 2017
Best Design Project (Thesis) Presenter · Asia Pacific College, 2016
Best Design Project (Thesis) Exhibitor · Asia Pacific College, 2016
Quizzer · Institute of Electronics Engineers of the Philippines (IECEP), 2014 – 2016
Honor Student · Asia Pacific College, 2012 – 2014
Champion, Inter-College Quiz Bee · Asia Pacific College, 2014
Top 4, Beat the Whiz (Math Quiz Bee) · Asia Pacific College, 2013

COMMUNITY ENGAGEMENTS

Google Developer Community <i>Google for Developers</i>	2022 – Present
Language Exchange Buddy Program <i>Sunchon National University</i>	2022 – 2023
Halalan 2019 Vote Validation <i>Parish Pastoral Council for Responsible Voting (PPCRV)</i>	2019